120 Days of Quantitative Finance

Month 3: Machine Learning and Deep Learning

Week 12: Deep Learning

Day 79: Activation Functions and Network Architecture

1 Introduction

Neural networks rely on two key components:

- Activation functions
- Network architecture

Activation functions introduce nonlinearity, while architecture determines how information flows through the network.

Both are critical for modeling complex relationships in financial data.

2 Why Activation Functions are Needed

Without activation functions, a neural network reduces to a linear model:

$$\hat{y} = W_2(W_1 x + b_1) + b_2$$

This can be simplified to a single linear transformation.

Therefore, nonlinear activation functions are necessary to model complex patterns.

3 Common Activation Functions

3.1 Sigmoid Function

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Properties:

- Output range: $(0, 1)$
- Smooth and differentiable
- Used in binary classification

3.2 Tanh Function

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Properties:

- Output range: $(-1, 1)$
- Zero-centered

3.3 ReLU (Rectified Linear Unit)

$$ReLU(x) = \max(0, x)$$

Properties:

- Computationally efficient
- Helps mitigate vanishing gradient problem
- Most widely used activation function

3.4 Leaky ReLU

$$LeakyReLU(x) = \begin{cases} x, & x > 0 \\ \alpha x, & x \leq 0 \end{cases}$$

This avoids the "dead neuron" problem in standard ReLU.

4 Network Architecture

A neural network is composed of multiple layers:

- Input layer
- Hidden layers
- Output layer

For a deep network with L layers:

$$a^{(l)} = \phi(W^{(l)} a^{(l-1)} + b^{(l)})$$

where:

- $l = 1, 2, ..., L$
- $a^{(0)} = x$ (input)
- $\phi(\cdot)$ is the activation function

5 Depth vs Width

Neural networks can vary in:

- Depth (number of layers)
- Width (number of neurons per layer)

Deep networks (more layers):

- Capture hierarchical patterns
- Better for complex data

Wide networks:

- Capture simpler relationships
- Easier to train

6 Universal Approximation Theorem

A key theoretical result states that a neural network with at least one hidden layer can approximate any continuous function:

$$f(x) \approx \hat{f}(x)$$

given sufficient neurons.

This explains why neural networks are so powerful.

7 Overfitting and Regularization

Complex architectures can lead to overfitting.

Common solutions include:

- Dropout
- Regularization (L1/L2)
- Early stopping

8 Applications in Finance

Neural network architectures are used in:

- Return prediction
- Volatility modeling
- Risk management
- Algorithmic trading

9 Conclusion

Activation functions enable neural networks to model nonlinear relationships, while architecture determines model capacity and flexibility.

Choosing the right combination is critical for building effective models in finance.